Acceleration, Offload, et al.

What's the deal?

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Question

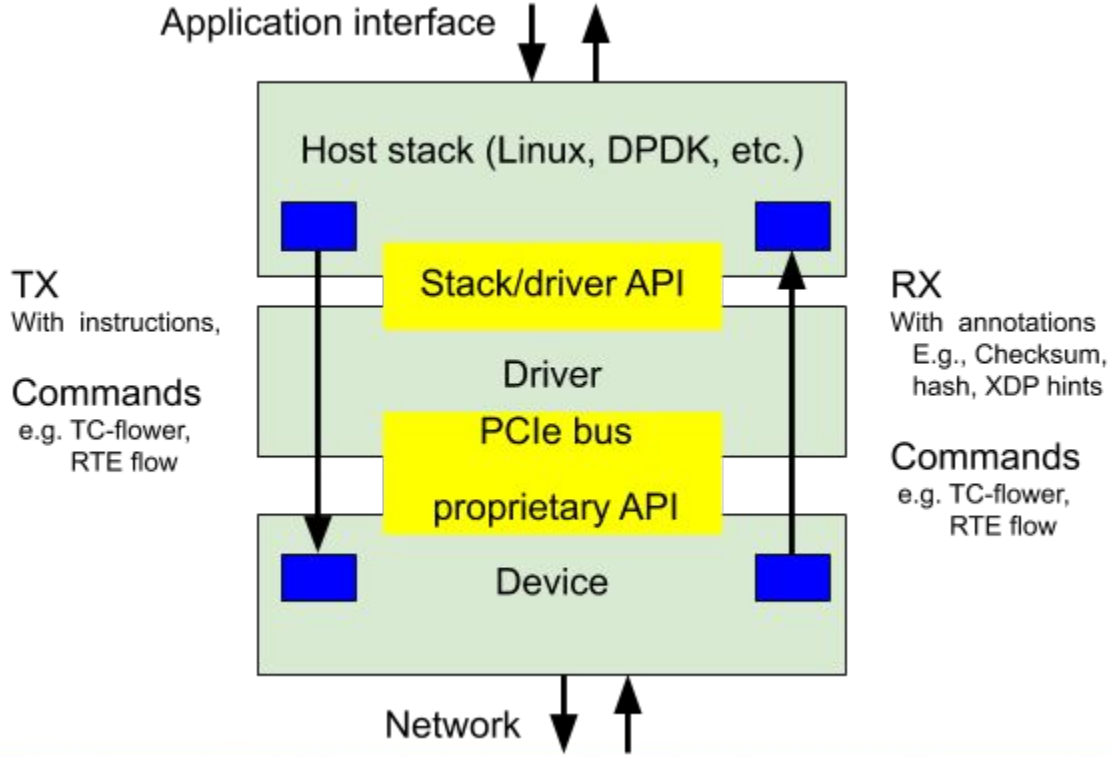
Why have so few offloads become ubiquitous
We got TX/RX checksum offload, RSS, TSO, maybe LRO on a good day.

IPsec, TLS, TC-flower, switchdev, some OVS kind of and sort of

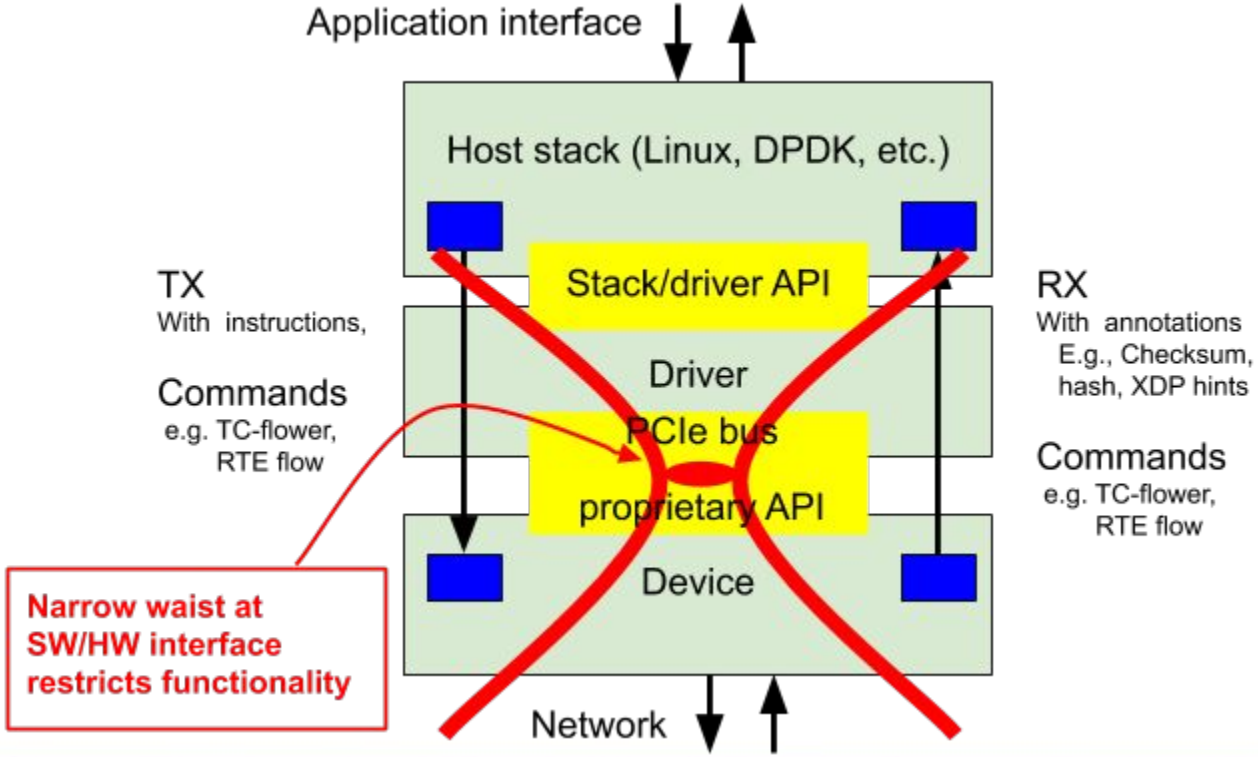
Acceleration versus offload

- **Offload**: Host stack requests additional operations when transmitting packets, device autonomously provided ancillary information with receive packet Forces offloads as opposed to acceleration, leads to CPU versus offload. E.g. csum, TSO
- **Acceleration**: Host stack invoke hardware accelerators in line in data path program. E.g. AES ins., GPU matrix operation acceleration

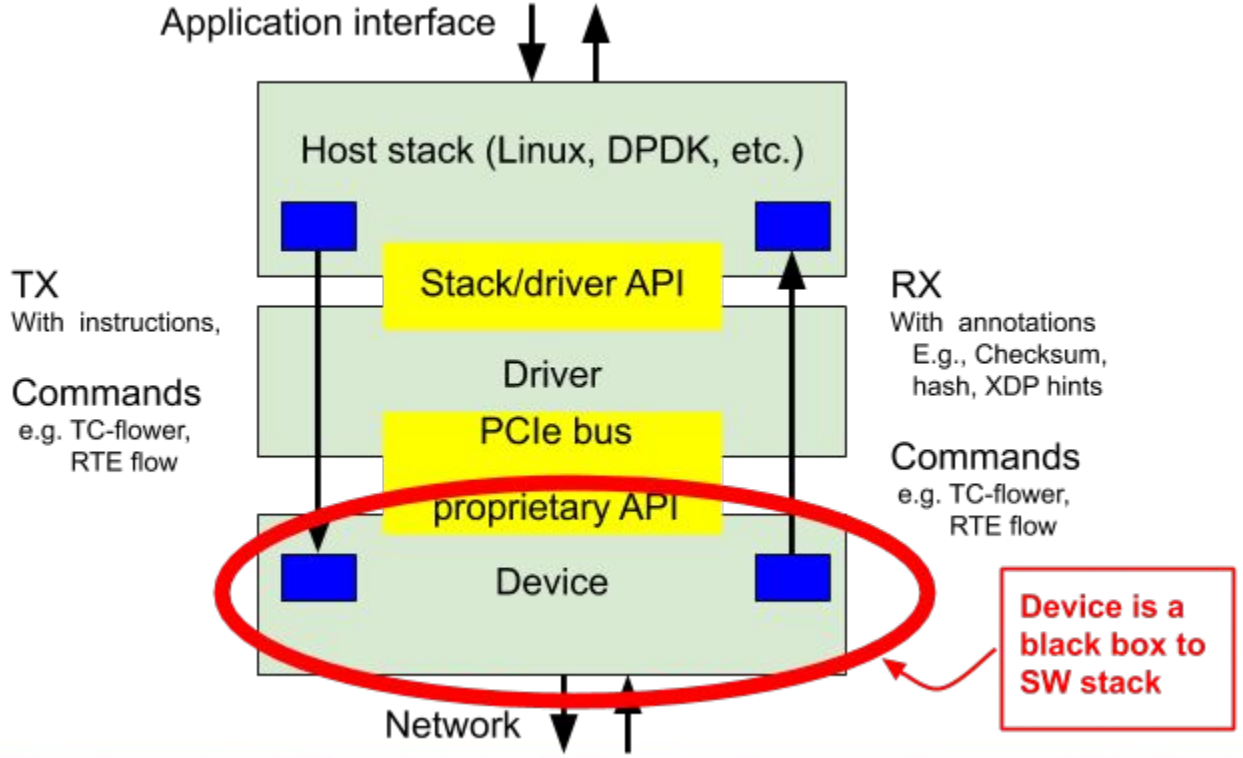
Traditional stack/driver/device model



The “narrow waist” problem



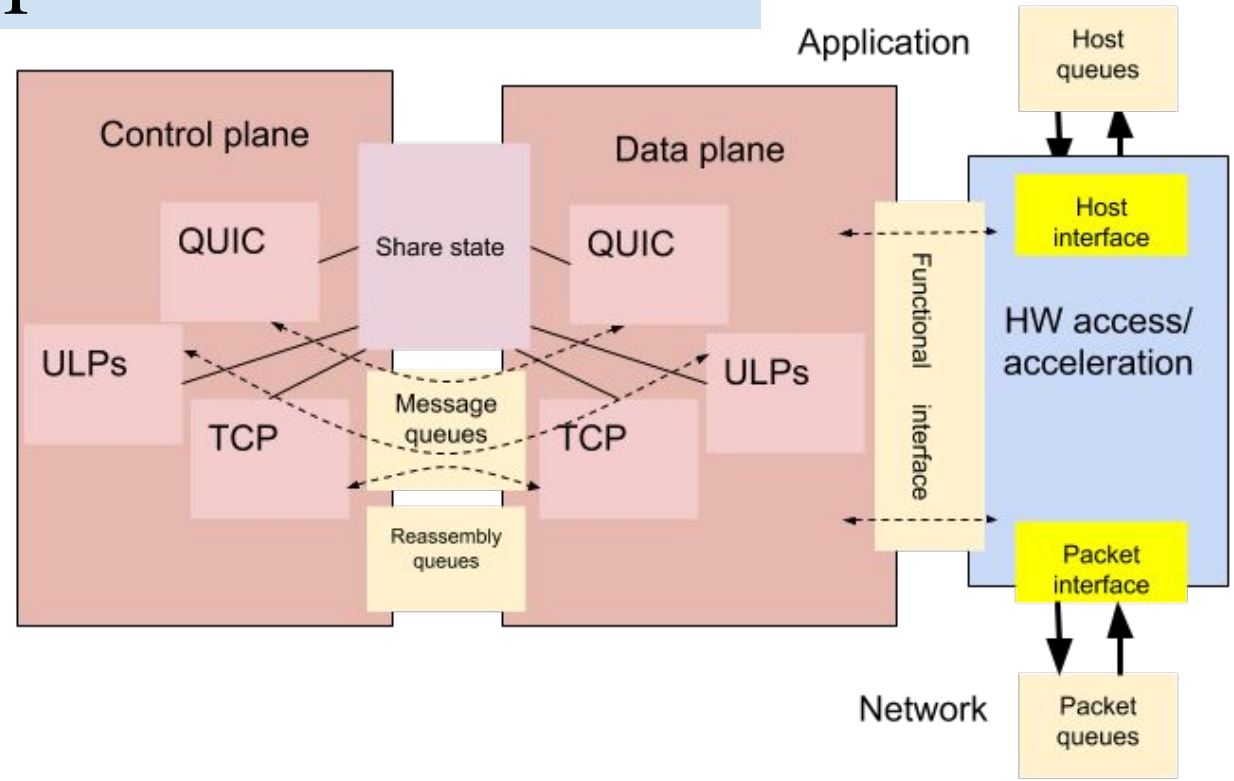
The “Black box” problem



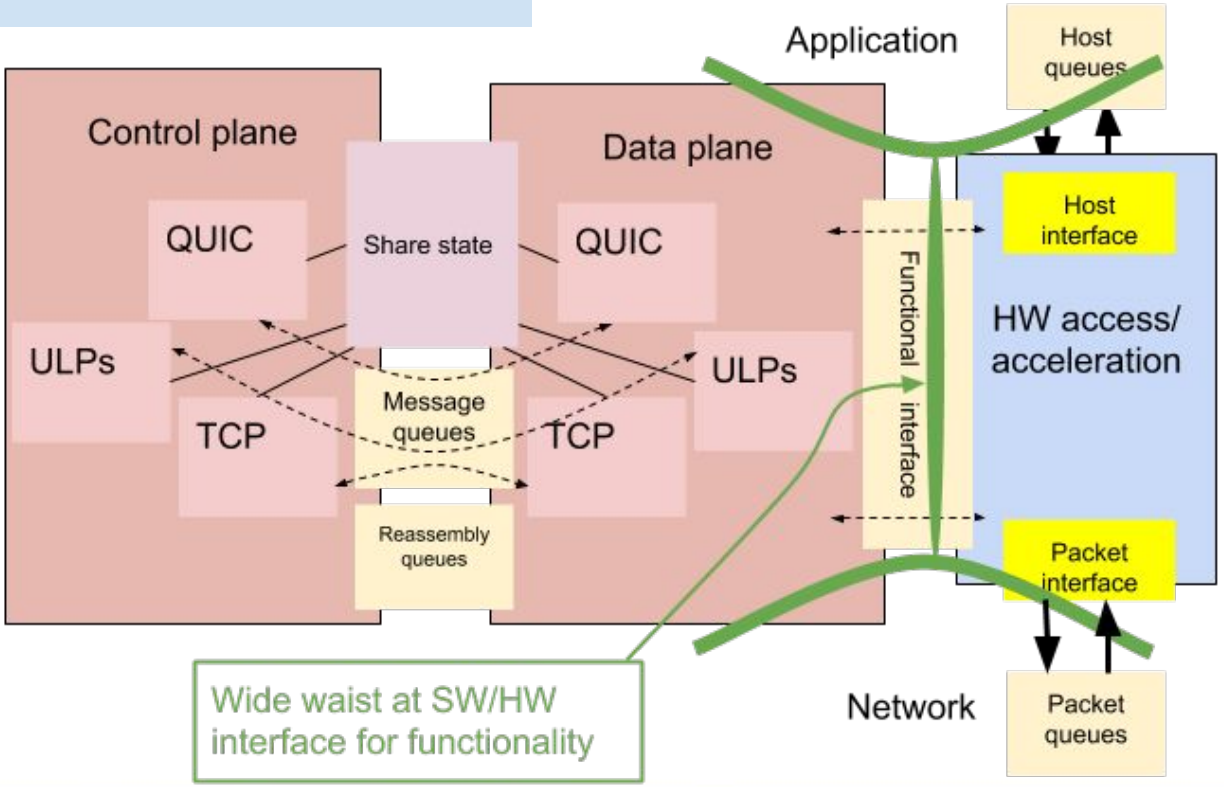
Outcome of this model

- Forces offloads as opposed to acceleration, leads to CPU versus offload (e.g. AVX-256 in OVS)
- Maintenance of shadow state kills any effort for stateless offload (e.g. TLS offload)
- Only a handful of offloads are ubiquitous, and even those are elusive to get right (e.g. csum offload, hash)
- Drives LCD with various OSes (e.g. bits in RX desc. Windows needs and Linux doesn’t)

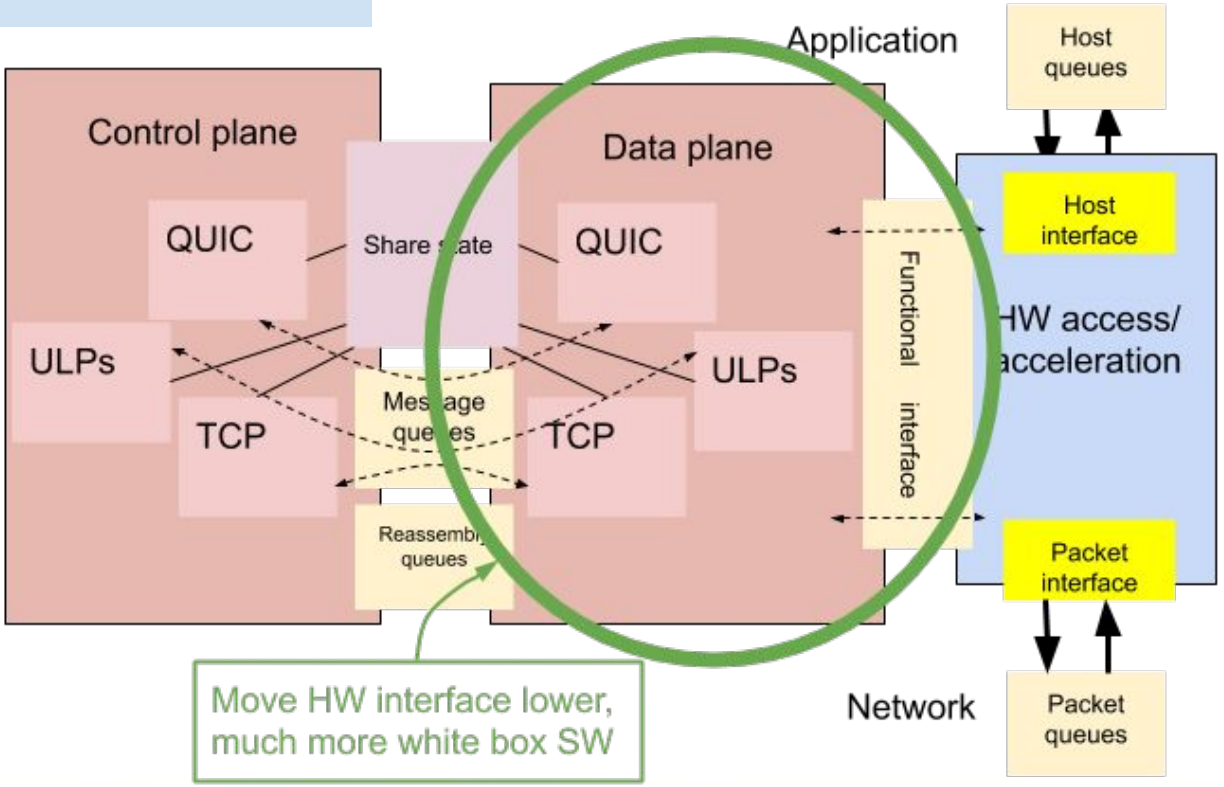
Split-plane acceleration



Wide waist at HW



Big white box



End slide